Reflection: The Pandemic Housing Boom and What Came After

Domain 3: Home Values

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Tableau Public Link:
https://public.tableau.com/app/profile/hrishad.amre/viz/HrishadAmre-DataVisualization/Story1?publish=yes

Iteration

My first draft showed the right data but didn't really say anything. The titles were labels ("Which Housing Markets Stayed Resilient?"), the boom and post-2022 growth charts lived on their own, and I had left colors and legends at Tableau's defaults. After my professor's feedback and the Baseline Visualization Standards, I rebuilt each dashboard around a single message and wrote the titles as the takeaway, for example "Nearly every county boomed; since 2022, 23% have lost value." The story now walks through three steps: "The boom was broad," "The aftermath wasn't," and "Rates made buying harder." The biggest changes were putting the boom and post-2022 histograms on top of each other so the leftward shift is visible at a glance, swapping the filled map for an equal-size tile map so small eastern states aren't lost, using a color-blind-safe blue and orange palette, and letting a colored subtitle act as the legend on the scatter. I also checked that each view does a different job (trend, distribution, place, affordability) instead of repeating the same point.

Challenge

The hardest part was the payment breakdown on the last dashboard. I wanted to show how much of the jump in monthly payment came from higher prices and how much from higher mortgage rates, but the answer depends on which one you change first: prices-first gives rates about 36% of the increase, rates-first gives about 54%. I didn't want to pick the flattering number, so I averaged both orders, which lands at roughly 55% prices and 45% rates, and I say so on the dashboard. A smaller but sneaky problem was missing data. 67 counties have no values, and my original segment calculation quietly labeled them "Persistent Lag." I caught it when the counts didn't add up, added a null check, and now note the exclusion in the footer.

Advanced techniques

Level of Detail (FIXED) expressions. Each county's Jan 2020, Jun 2022 and Jun 2026 values, the median cutoffs behind the four segments, and the price-versus-rate split all use FIXED calculations, so the numbers stay put no matter what else is in the view.

A parameter-driven custom view. A "Show Segment" dropdown recalculates each state's share of counties in the chosen segment and recolors a tile map I built from a hand-coded grid of state positions.

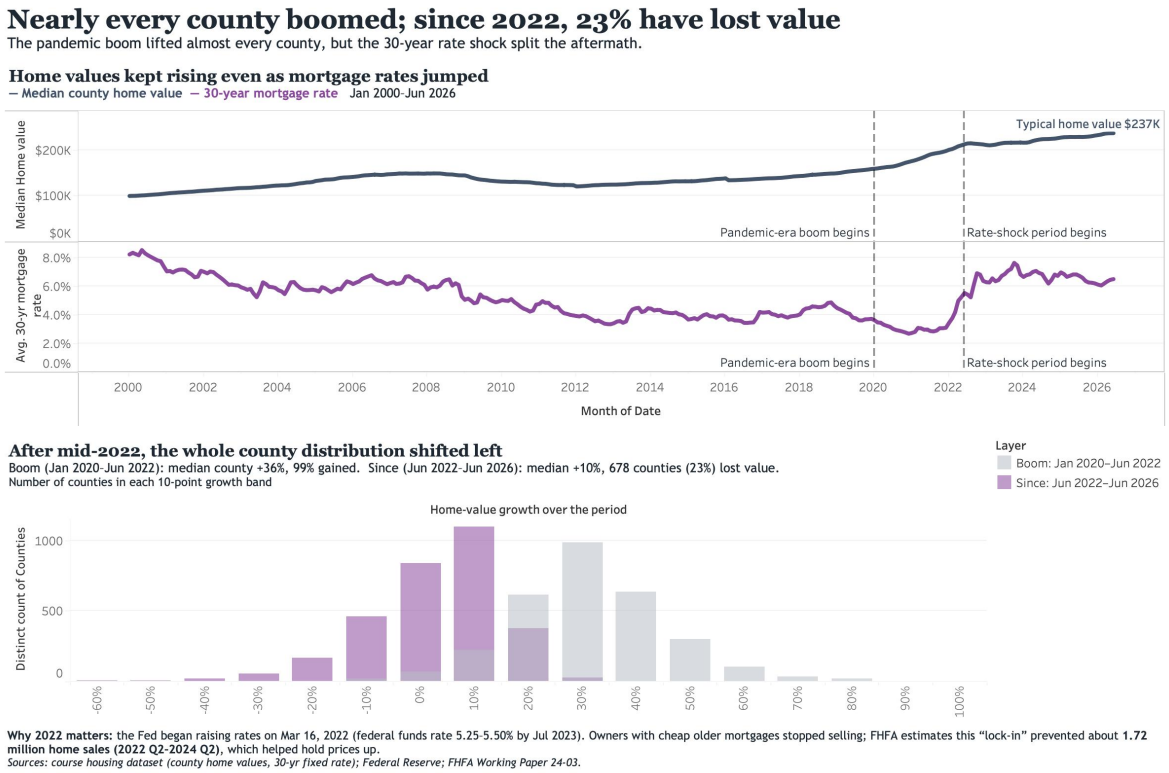
A dashboard action. Selecting dots in the scatter filters the county-type bar chart, so you can see where a group of counties sits.

Calculated aggregates such as the share of counties that lost value and median payment-to-income by state.

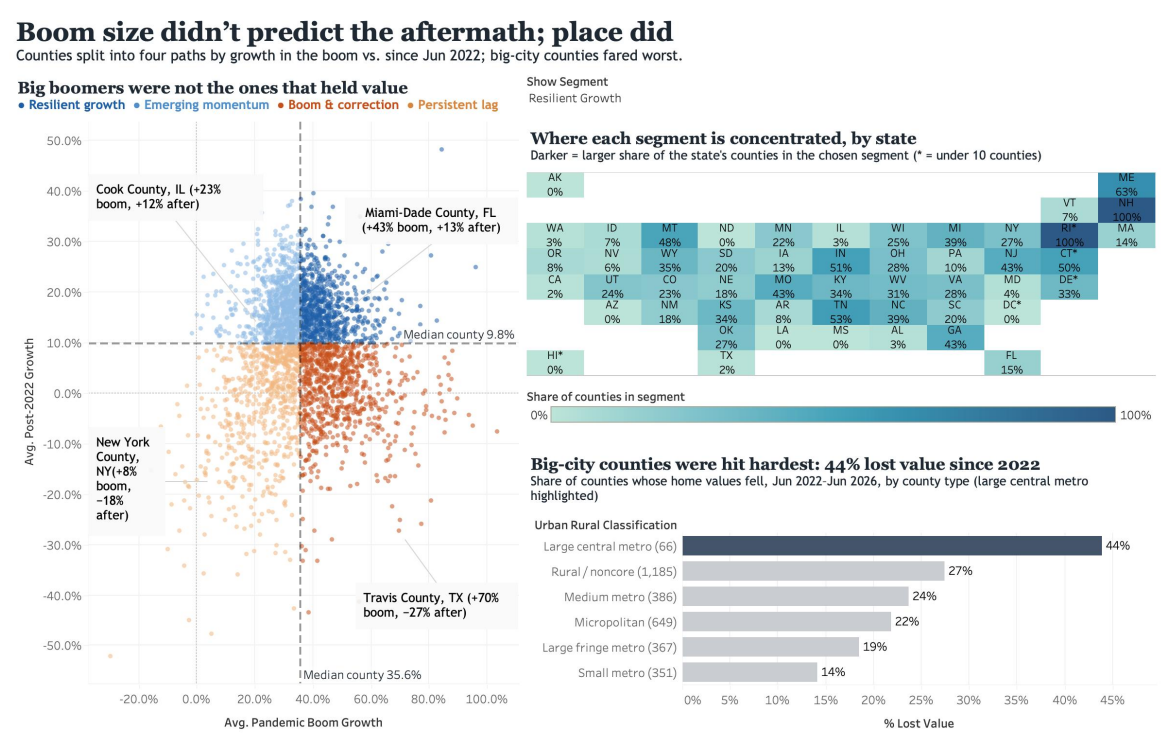
External context

Three outside sources shaped the story. The Federal Reserve began raising rates on March 16, 2022. A Federal Housing Finance Agency paper (Working Paper 24-03) estimates the mortgage "lock-in" effect kept about 1.72 million homes off the market between 2022 Q2 and 2024 Q2, which helps explain why prices held up. And a Congressional Research Service report on housing cost burdens gives the 30%-of-income threshold I use to flag unaffordable states.

Appendix: The three dashboards in the story



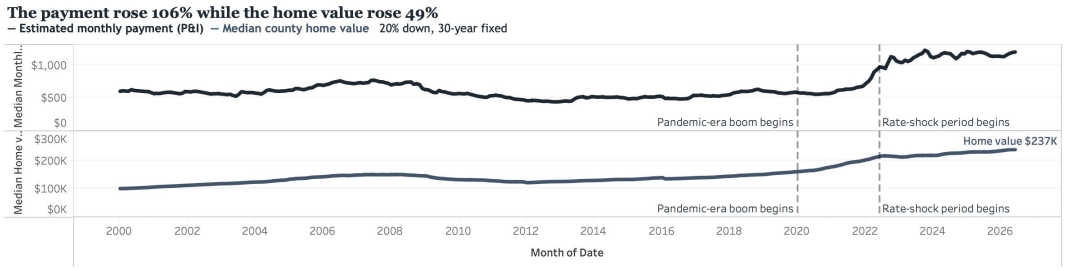
DB1: nearly every county boomed; since 2022, 23% have lost value

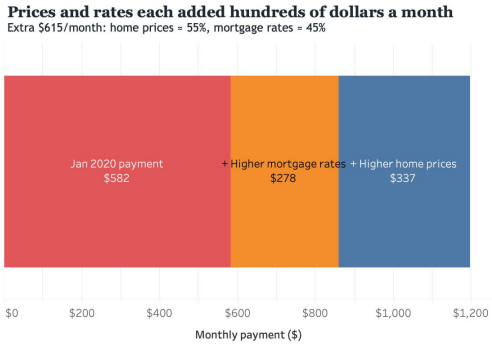


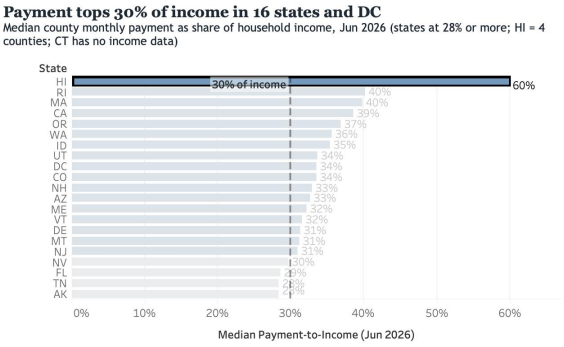
DB2: boom size did not predict the aftermath; place did

Higher rates, not just prices, doubled the monthly payment

Estimated payment on a typical home rose 106% while home values rose 49%; burdens are heaviest in the West and Northeast.







Method: principal & interest, 20% down, 30-year fixed at that month's average rate. Payment-to-income = annual P&I ÷ county median income; 30% = standard cost-burden line (CRS, 2024).

DB3: higher rates, not just prices, doubled the monthly payment